

LASER

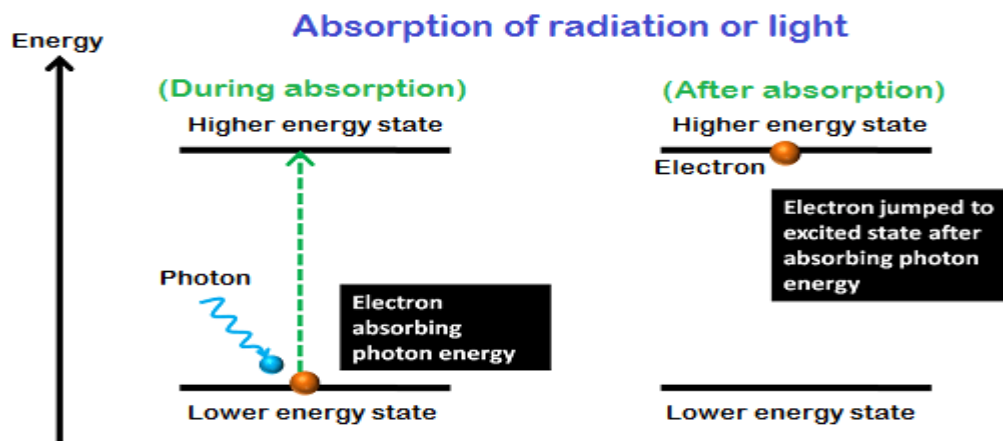
It is a device to produce a highly intense monochromatic narrow beam of light in which the waves are convergent. Laser is an acronym for light amplification by stimulated emission of radiation. The light emitted from the conventional light source (eg: sodium lamp, candle) is said to be incoherent. Because the radiation emitted from different atom do not have any definite phase relationship with each other. Lasers are much important because the light sources having high monochromaticity, high intensity, high directionality and high coherence.

Laser principle now a days is extended up to γ - rays hence Gamma ray lasers are called Grazers. The first two successful lasers developed during 1960 were Ruby laser and He- Ne lasers. Some lasers emit light in pulses while others emit as a continuous wave.

Stimulated Absorption:

The process of absorbing energy from photons is called absorption of radiation.

Let us consider that the energy level of ground state electrons or lower energy state electrons is E_1 and the next higher energy level or higher energy state is E_2 .



Physics and Radio-Electronics

When ground state electrons (E_1) absorb sufficient energy from photons, they jump into the next higher energy level or state (E_2). When the ground state electrons absorb energy which is equal to the energy difference between the two energy states ($E_2 - E_1$), the electrons jump from ground state (E_1) to the excited state or higher energy level (E_2). The electrons in the higher energy level are called excited electrons.

The light or photons energy applied to excite the electrons can be mathematically written as

$$h\nu = E_2 - E_1$$

Where h = Planck's constant

ν = Frequency of photon

E_1 = Lower energy level electrons or ground state electrons

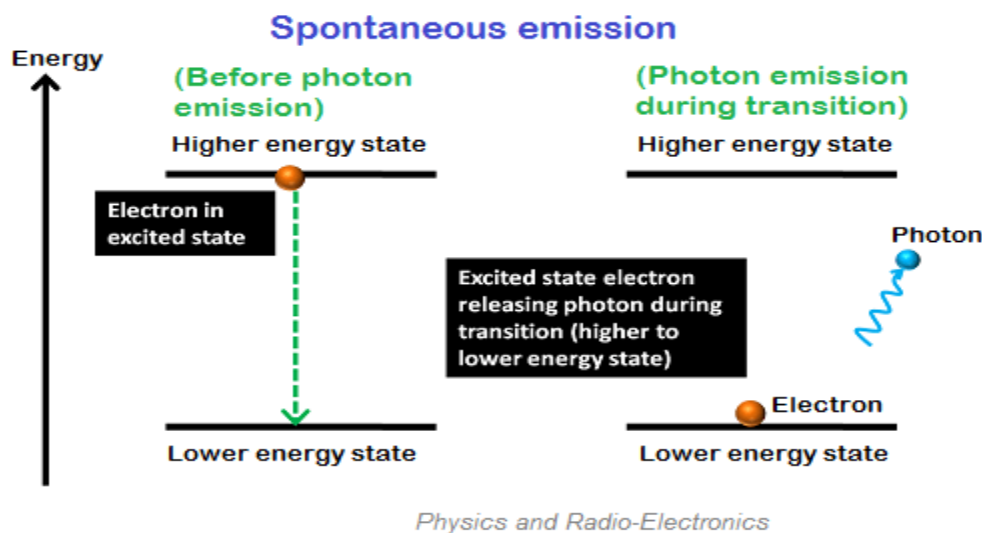
E_2 = Higher energy level electrons or excited state electrons

Absorption occurs only if the energy of photon exactly matches the difference in energy between the two electron shells or orbits.

Spontaneous Emission

The process by which excited electrons emit photons while falling to the ground level or lower energy level spontaneously is called spontaneous emission.

When the electrons in the ground state or lower energy state (E_1) absorb sufficient energy from photons, they jump to the excited state or next higher energy state (E_2). The electrons in the excited state do not stay for a long period because the lifetime of electrons in the higher energy state or excited state is very small, of the order of 10^{-8} sec. Hence, after a short period, they fall back to the ground state by releasing energy in the form of photons or light.

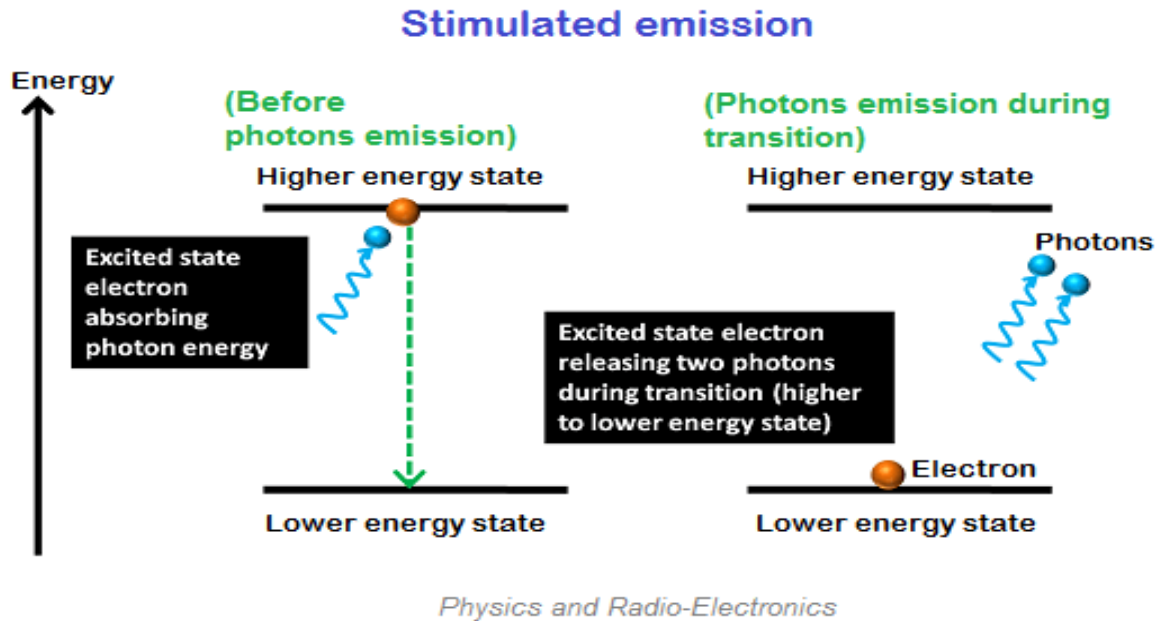


In spontaneous emission, the electrons changing from one state (higher energy state) to another state (lower energy state) occurs naturally. So the photon emission also occurs naturally or spontaneously. The photons emitted due to spontaneous emission do not flow exactly in the same direction of incident photons. They flow in the random direction.

Stimulated Emission:

The process by which electrons in the excited state are stimulated to emit photons while falling to the ground state or lower energy state is called stimulated emission.

Unlike the spontaneous emission, in this process, the light energy or photon energy is supplied to the excited electrons. In stimulated emission, the electrons in the excited state need not wait for natural spontaneous emission to occur. An alternative method is used to stimulate excited electron to emit photons and fall back to ground state.



The incident photon stimulates or forces the excited electron to emit a photon and fall into a lower state or ground state. The energy of a stimulating or incident photon must be equal to the energy difference between the two electron shells.

In this process, the excited electron releases an additional photon of same energy (same frequency, same phase, and in the same direction) while falling into the lower energy state. Thus, two photons of same energy are released while electrons falling into the ground state.

In stimulated emission process, each incident photon generates two photons.

The photons emitted in the stimulated emission process will travel in the same direction of the incident photon. Many ways exist to produce light, but the stimulated emission is the only method known to produce coherent light (beam of photons with the same frequency). All the photons in the stimulated emission have the same frequency and travel in the same direction.

Characteristics of Laser:

The four characteristics of a laser radiation over conventional light sources are

- (1) Laser is highly monochromatic
- (2) Laser is highly directional
- (3) Laser is highly coherent
- (4) The intensity of laser is very high

1. Highly monochromatic:

The band width of ordinary light is about $1000\text{\AA}$. The band width of laser light is about $10\text{\AA}$. The narrow band width of a laser light is called as high monochromaticity.

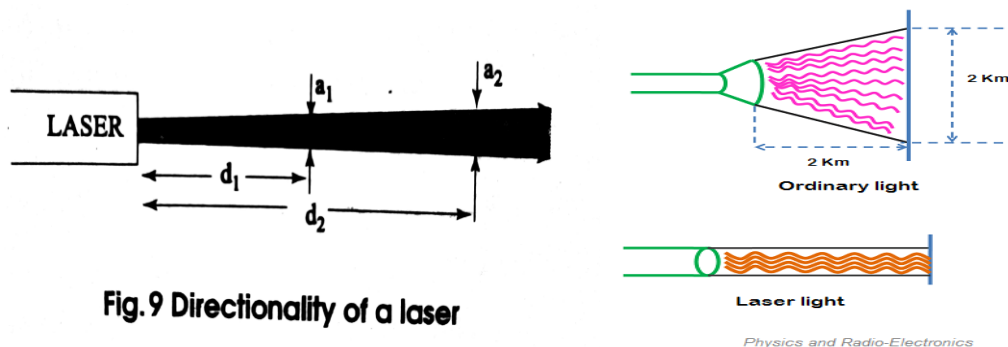
Band width: The spread of the wavelength (frequency) about the wavelength of maximum intensity is band width. Laser light is more monochromatic than that of a conventional light source. Because of this monochromaticity large energy can be concentrated in to an extremely small band width.

The degree of monochromaticity of light from any source can be defined in terms of line width (spectral width) of the source. The frequency spread $\Delta\nu$ of spectral line is related to the wavelength spread $\Delta\lambda$ as, For a white light source $\Delta\lambda = 300\text{nm}$ whereas for a gas discharge line $\Delta\lambda = 0.01\text{nm}$.

2. High directionality:

In conventional light sources (lamp, sodium lamp and torchlight), photons will travel in random direction. Therefore, these light sources emit light in all directions.

On the other hand, in laser, all photons will travel in same direction. Therefore, laser emits light only in one direction. This is called directionality of laser light. The width of a laser beam is extremely narrow. Hence, a laser beam can travel to long distances without spreading.



If an ordinary light travels a distance of 2 km, it spreads to about 2 km in diameter. On the other hand, if a laser light travels a distance of 2 km, it spreads to a diameter less than 2 cm. The directionality of laser beam is given by (or) expressed in divergence.

The divergence $\Delta\theta = (r_2 - r_1) / d_2 - d_1$ Where r_2 , r_1 are the radius of laser beam spots and d_2 , d_1 are distances respectively from the laser source. Hence for getting a high directionality there should be low divergence.

3. Coherence:

Laser has high degree of ordering than other common sources. Due to its coherence only it is possible to create high power (10^{12} watts) in space with laser beam of $1\mu\text{m}$ diameter. There are two independent concepts of coherence.

i) Spatial coherence ii) Temporal coherence

i. Spatial coherence: The two light fields at different point in space maintain a constant phase difference over any time (t) they are said to be spatial coherence or transverse coherence. When we choose two spatial points P_1 and P_2 on the cross section of the output beam, if they are in phase with one another or maintain same phase differences then they are spatially coherent.

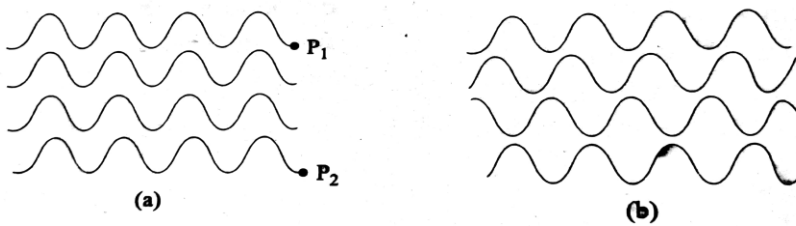
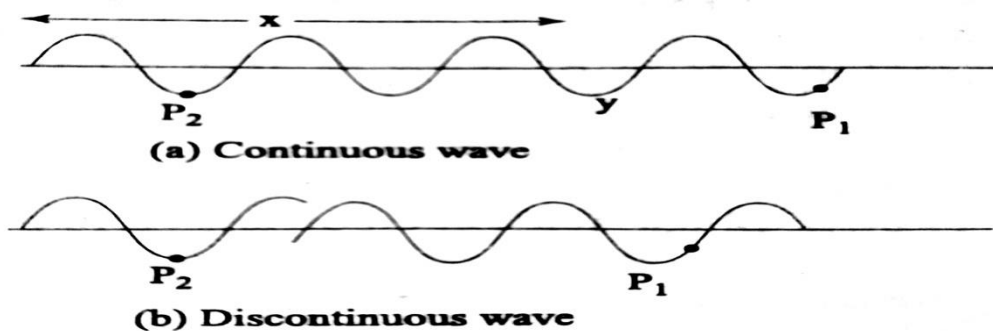


Fig. 12 (a) Spatially coherent waves (b) spatially incoherent wave

In He- Ne gas laser the coherence length (l_c) is about 600km. It means over the distance the phase difference is maintained over any time. For sodium light it is about 3cm.

ii. Temporal coherence: The correlation of amplitude and phase between any two points on a wave train over a period of time is called temporal coherence or longitudinal coherence. If the two points P_1 and P_2 which are on the same wave train carries same phase and amplitude then it is said to be temporal coherence.



For He- Ne laser it is about 10^{-3} second, for sodium it is about 10^{-10} second only.

4. High intensity:

Lasers are bright and intense light sources because of coherence and directionality. Intensity of a wave is the energy per unit time flowing through a unit area. The light from an ordinary source spreads out uniformly in all directions and form spherical wave fronts around it.

Ex:- If we look a 100W bulb from a distance of 30cm the power entering the eye is 1 / 1000 of watt. But in case of a laser light, energy is in small region of space and in a small wavelength and hence is said to be of great intensity. The power range of laser is about 10^{-3} W for gas laser and 10^9 W for solid state laser.

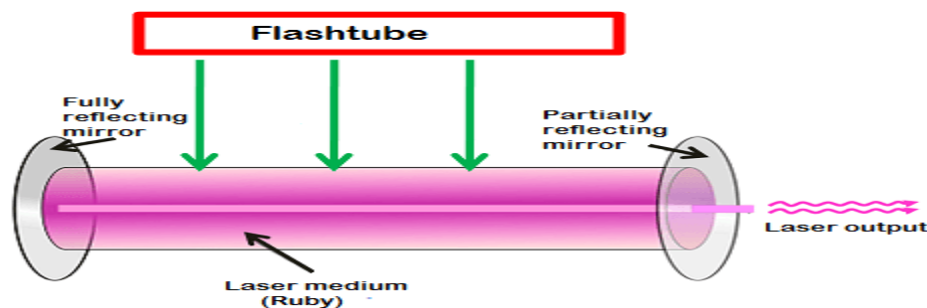
Optical Resonator:

The laser medium is surrounded by two parallel mirrors which provides feedback of the light. One mirror is fully reflective (100 % reflective) whereas another one is partially reflective (<100 % reflective). These two mirrors as a whole is called optical resonator. Optical resonator is also known as optical cavity or resonating cavity.

These two mirrors are given optical coatings which determine their reflective properties. Optical coating is a thin layer of material deposited on materials such as mirror or lens. Each mirror is coated differently. Therefore, each mirror will reflect the light differently. One mirror will completely reflect the light whereas another one will partially reflect the light.

The completely reflective mirror is called high reflector whereas the partially reflective mirror is called output coupler. The output coupler will allows some of the light to leave the optical cavity to produce the laser's output beam.

When energy is supplied to the laser medium, the lower energy state electrons in the laser medium will moves to excited state. After a short period, the electrons in the excited state will fall back to the ground state by releasing energy in the form of photons or light. This process of emission of photons is called spontaneous emission. Thus, light is produced in an active medium by a process called spontaneous emission.



The light generated within the laser medium will bounce back and forth between the two mirrors. This stimulates other electrons to release light while falling to the ground state. Likewise, a large number of electrons are stimulated to emit light. Thus, optical gain is achieved. This amplified light escapes through the partially reflecting mirror. The process of stimulating electrons of other atoms to produce light in the laser medium is called stimulated emission.

The light in the laser medium is reflected many hundreds of times between the mirrors before it escape through the partially reflecting mirror. The light escaped from the partially reflecting mirror is produced by the stimulated emission process. Hence, this light will travel to large distances without spreading in the space.

Meta stable state: Metastable state is particular excited state of an atom, nucleus, or other system that has a longer lifetime than the ordinary excited states and that generally has a shorter lifetime than the lowest, often stable, energy state, called the ground state. A metastable state may thus be considered a kind of temporary energy trap or a somewhat stable intermediate stage of a system where the life time of atom ,nucleus or system is in 10^{-6} Sec.

Active Medium:

The laser medium is a medium where spontaneous and stimulated emission of radiation takes place.

After receiving sufficient energy from source, the electrons in the lower energy state or ground state are excited to the higher energy state (in the laser medium) and spontaneous emission will takes place. In spontaneous emission, each electron emits a single photon while falling to the ground state.

When these emitted photons collide with the electrons in the excited state or meta stable state, it forces meta stable electrons to fall back to the ground state. As a result, electrons again release energy in the form of photons. This is called stimulated emission. In stimulated emission, each electron emits two photons while falling to the ground state. When these emitted photons are again interacted with the meta stable state electrons then again two photons are emitted by each electron. Thus, millions of photons are generated by using only a small number of photons. Laser medium is also known as active medium or gain medium.

Population Inversion:

The number of atoms in higher energy level is more than the number of atoms in lowest energy level i.e. $N_2 > N_1$. The process of making of higher population in higher energy level than the population in lower energy level is known as population inversion.

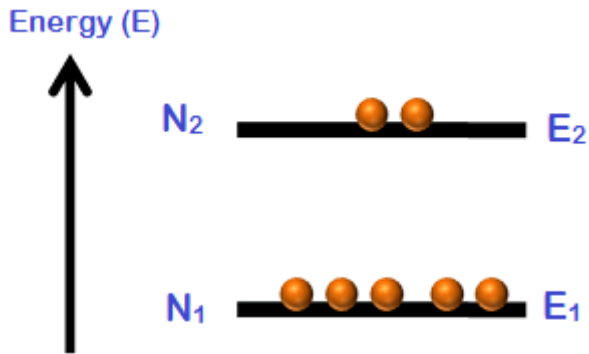
Consider a group of electrons with two energy levels E_1 and E_2 .

E_1 is the lower energy state and E_2 is the higher energy state.

N_1 is the number of electrons in the energy state E_1 .

N_2 is the number of electrons in the energy state E_2 .

The number of electrons per unit volume in an energy state is the population of that energy state.



Population inversion cannot be achieved in a two energy level system. Under normal conditions, the number of electrons (N_1) in the lower energy state (E_1) is always greater as compared to the number of electrons (N_2) in the higher energy state (E_2) $N_1 > N_2$

Population inversion is achieved by pumping the atoms from the ground level to the higher energy level.

Pumping Schemes: There are some restrictions imposed on two level laser system. To overcome such restrictions three and four level laser systems are introduced.

Optical pumping: In this method, light is used to supply energy to the laser medium. An external light source like xenon flash lamp is used to produce more electrons (a high population) in the higher energy level of the laser medium.

Electrical discharge: In this method of pumping, electric discharge acts as the pump source or energy source. A high voltage electric discharge (flow of electrons, electric charge, or electric current) is passed through the laser medium or gas. This method of pumping is used in gas lasers such as argon lasers.

Chemical reaction: If an atom or a molecule is produced through some chemical reaction and remains in an excited state at the time of production, then it can be used for pumping.

Thermal Pumping: In thermal pumping, heat acts as the pump source or energy source. In this method, population inversion is achieved by supplying heat into the laser medium.

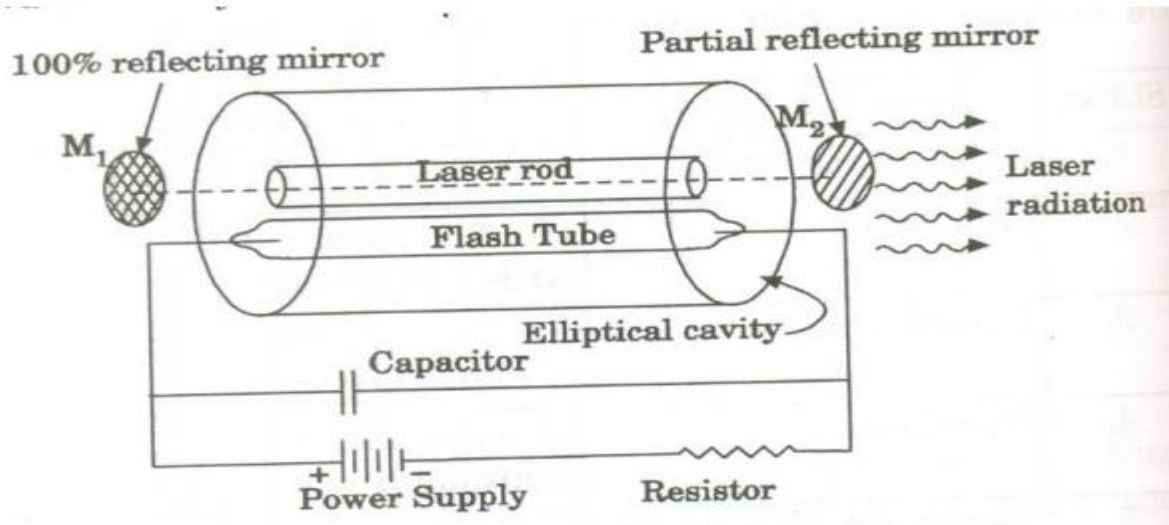
Injection current etc.

Nd:YAG Laser:

Nd: YAG laser is a neodymium based laser. Nd stands for Neodymium (rare earth element) and YAG stands for Yttrium Aluminum Garnet ($Y_3Al_5O_{12}$). It is a four level solid state laser.

Principle:

The active medium Nd: YAG rod is optically pumped by Krypton flash tubes. The Neodymium ions (Nd^{3+}) are raised to excited levels. During the transition from metastable state to ground state, a laser beam of wavelength $1.064\mu m$ is emitted.



Construction:

The construction of Nd: YAG laser is as shown in the figure . A small amount of Yttrium ions (Y^{3+}) is replaced by Neodymium (Nd^{3+}) in the active element of Nd: YAG crystal.

This active element is cut into a cylindrical rod. The ends of the cylindrical rod are highly polished and they are made optically flat and parallel. This cylindrical rod (laser rod) and a pumping source (flash tube) are placed inside a highly (reflecting) elliptical reflector cavity.

The optical resonator is formed by using two external reflecting mirrors. One mirror (M₁) is 100% reflecting while the other mirror (M₂) is partially reflecting.

Characteristics:

Type: It is a four level solid state laser.

Active medium: The active medium is Nd: YAG laser.

Pumping method: Optical pumping is employed for pumping action.

Pumping source: Xenon or Krypton flash tube is used as pumping source.

Optical resonator: Two ends of Nd: YAG rod is polished with silver (one end is fully silvered and the other is partially silvered) are used as optical resonator.

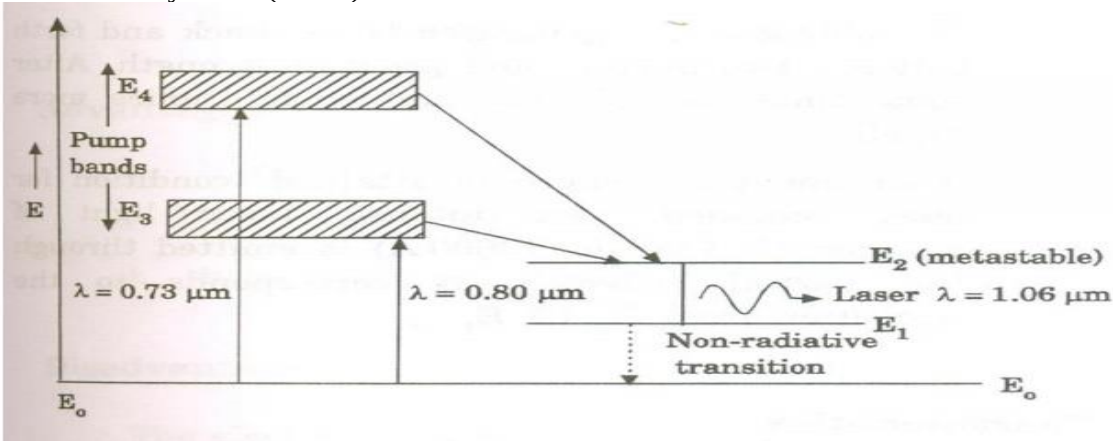
Power output: The power output is approximately 70 watt.

Nature of output: The nature of output is pulsed or continuous beam of light.

Wavelength of the output: The wavelength of the output beam is $1.06\mu\text{m}$ (infra-red).

Working:

Figure 2.18 shows the energy level diagram for Nd: YAG laser. These energy levels are those of Neodymium (Nd^{3+}) ions.



1. When the krypton flash lamp is switched on, by the absorption of light radiation of wavelength $0.73\mu\text{m}$ and $0.8\mu\text{m}$, the Neodymium(Nd^{3+}) atoms are raised from ground level E_0 to upper levels E_3 and E_4 (Pump bands).
2. The Neodymium ions atoms make a transition from these energy levels E_2 by non-radiative transition. E_2 is a metastable state.
3. The Neodymium ions are collected in the level E_2 and the population inversion is achieved between E_2 and E_1 .
4. An ion makes a spontaneous transition from E_2 to E_1 , emitting a photon of energy $h\nu$. This emitted photon will trigger a chain of stimulated photons between E_2 and E_1 .
5. The photons thus generated travel back and forth between two mirrors and grow in strength. After some time, the photon number multiplies more rapidly.
6. After enough strength is attained (condition for laser being satisfied), an intense laser light of wavelength $1.06\mu\text{m}$ is emitted through the partial reflector. It corresponds to the transition from E_2 to E_1 .

Advantages:

1. It has high energy output.
2. It has very high repetition rate operation

3. It is much easy to achieve population inversion.

Disadvantages:

The electron energy level structure of Nd^{3+} in YAG is complicated.

Applications:

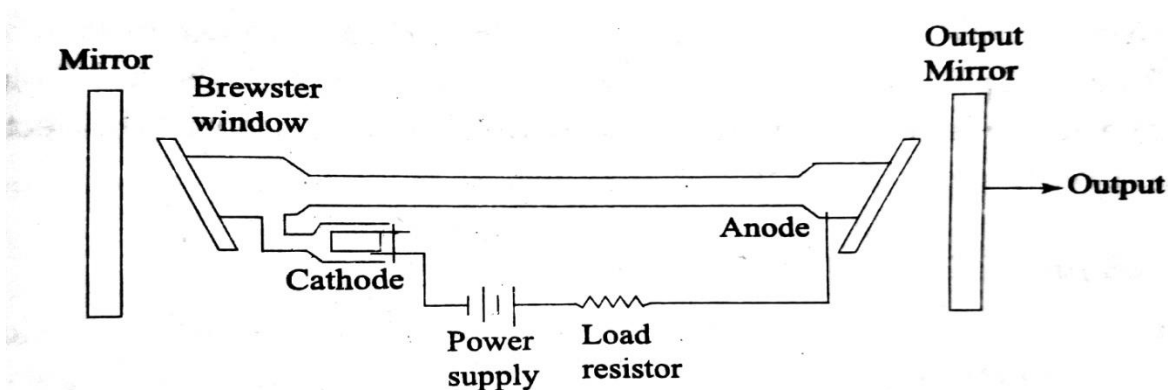
1. It finds many applications in range finders and illuminators.
2. It is widely used in engineering applications such as resistor, trimming scribing, micro machining operations as well as welding, drilling etc.
3. It finds many medical applications such as endoscopy, urology, neurosurgery, ENT, gynecology, dermatology, dental surgery and general surgery.

He – Ne laser:

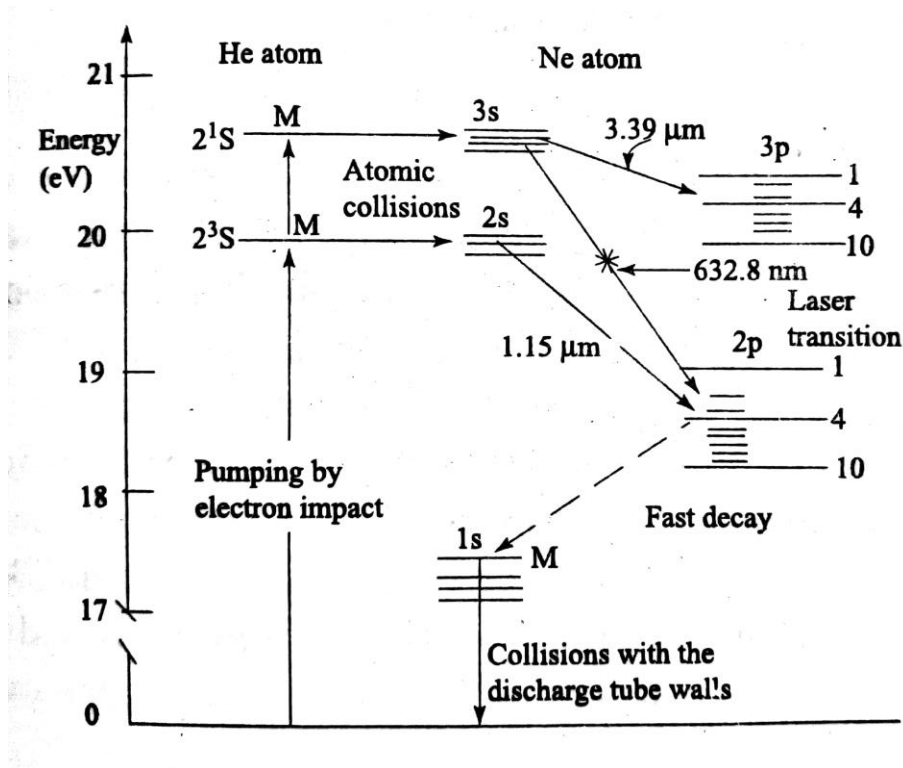
The helium-neon laser consists of three essential components:

- Pump source (high voltage power supply)
- Gain medium (laser glass tube or discharge glass tube)
- Resonating cavity

Construction: He - Ne gas laser consists of a gas discharge tube of length 80cm and diameter of 1cm. The tube is made up of quartz and is filled with a mixture of Neon under a pressure of 0.1mm of Hg and Helium under the pressure of 1mm of Hg. The ratio of He-Ne mixture of about 10:1, hence the no of helium atoms are greater than neon atoms. The mixtures is enclosed between a set of parallel mirrors forming a resonating cavity, one of the mirror is completely reflecting and the other is partially reflecting in order to amplify the output laser beam.



Working: When an electric discharge is passed through the gaseous mixture electrons are accelerated down the tube, these accelerated electrons collide with the helium atoms and excite them to higher energy levels since the levels are meta stable energy levels of He. He atoms spend sufficiently long time and collide with Neon atoms in the ground level E_1 . Then Neon atoms are excited to the higher energy levels 2S & 3S and helium atoms are de excited to the ground energy level.



Since 3S & 2S of Ne atom are metastable states, population inversion takes place at these levels. The stimulated emission takes place between 3S to 2P gives a laser light of wavelength 632.8nm ($6328\text{\AA}$) and the stimulated emission between 3S and 3P gives a laser light of wave length 3.39 μm . Another stimulated emission between 2S to 2P gives a laser light of wave length 1.15 μm . The Neon atoms undergo spontaneous emission from 2P to 1S and 3P to 1S. Finally the Neon atoms are returned to the ground state from 1S by non-radiative diffusion and collision process. After arriving at the ground state, once again the

Neon atoms are raised to 3S & 4 by excited Helium atoms thus we can get continuous output from He-Ne laser.

Molecular Gas Laser

In a molecular gas laser, laser action is achieved by transitions between vibrational and rotational levels of molecules. Its construction is simple and the output of this laser is continuous.

In CO_2 molecular gas laser, transition takes place between the vibrational states of Carbon dioxide molecules.

CO_2 Molecular Gas Laser

It was the first molecular gas laser developed by Indian born American scientist Prof.C.K.N.Pillai.

It is a four level laser and it operates at $10.6\ \mu m$ in the far IR region. It is a very efficient laser.

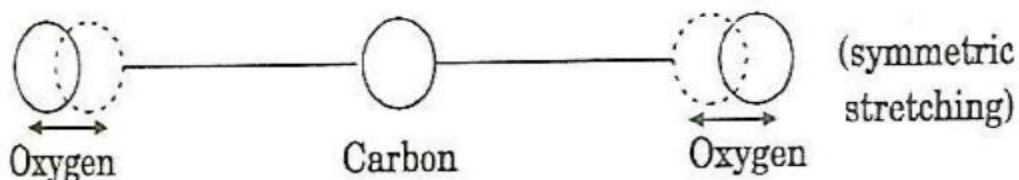
Energy States of CO_2 Molecules

A carbon dioxide molecule has a carbon atom at the center with two oxygen atoms attached, one at both sides. Such a molecule exhibits three independent modes of vibrations. They are

- a) Symmetric stretching mode.
- b) Bending mode
- c) Asymmetric stretching mode.

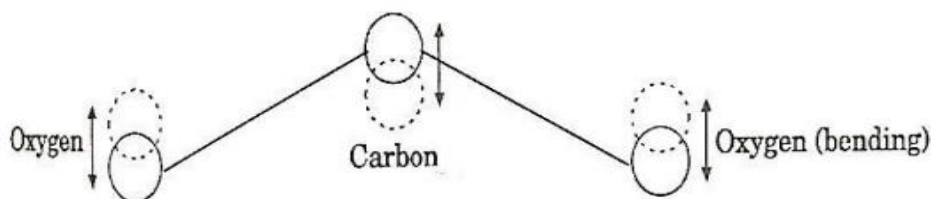
a. Symmetric Stretching Mode

In this mode of vibration, carbon atoms are at rest and both oxygen atoms vibrate simultaneously along the axis of the molecule departing or approaching the fixed carbon atoms.

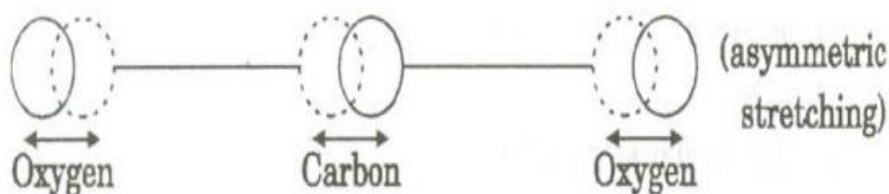


b. Bending Mode:

In this mode of vibration, oxygen atoms and carbon atoms vibrate perpendicular to molecular axis.



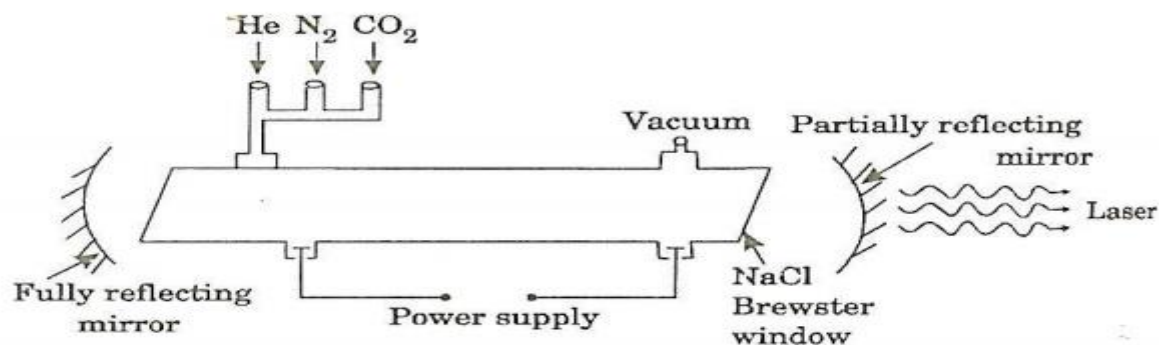
c. Asymmetric Stretching Mode:



In this mode of vibration, oxygen atoms and carbon atoms vibrate asymmetrically, i.e., oxygen atoms move in one direction while carbon atoms in the other direction.

Principle:

The active medium is a gas mixture of CO_2 , N_2 and He. The laser transition takes place between the vibrational states of CO_2 molecules.



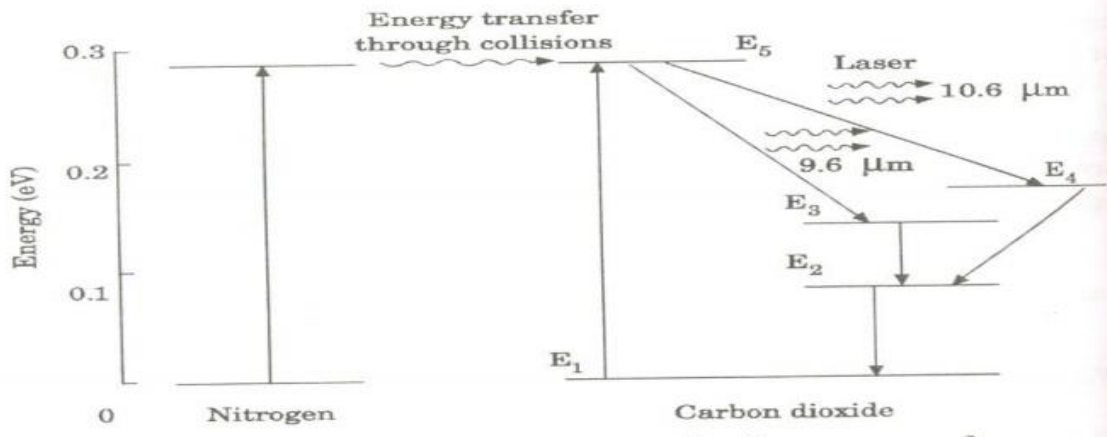
Construction:

It consists of a quartz tube 5 m long and 2.5 cm in the diameter. This discharge tube is filled with gaseous mixture of CO_2 (active medium), helium and nitrogen with suitable partial pressures. The terminals of the discharge tubes are connected to a D.C power supply. The ends of the discharge tube are fitted with NaCl Brewster windows so that the laser light generated will be

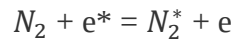
polarized. Two concave mirrors one fully reflecting and the other partially form an optical resonator.

Working:

Figure shows energy levels of nitrogen and carbon dioxide molecules.



When an electric discharge occurs in the gas, the electrons collide with nitrogen molecules and they are raised to excited states. This process is represented by the equation



N_2 = Nitrogen molecule in ground state e^* = electron with kinetic energy

N_2^* = nitrogen molecule in excited state e = same electron with lesser energy

Now N_2 molecules in the excited state collide with CO_2 atoms in ground state and excite to higher electronic, vibrational and rotational levels.

This process is represented by the equation $N_2^* + CO_2 = CO_2^* + N_2$

N_2^* = Nitrogen molecule in excited state. CO_2 = Carbon dioxide atoms in ground state

CO_2^* = Carbon dioxide atoms in excited state N_2 = Nitrogen molecule in ground state.

Since the excited level of nitrogen is very close to the E_5 level of CO_2 atom, population in E_5 level increases.

As soon as population inversion is reached, any of the spontaneously emitted photon will trigger laser action in the tube. There are two types of laser transition possible.

1. Transition E_5 to E_4 :

This will produce a laser beam of wavelength $10.6\mu m$

2. Transition E_5 to E_3

This transition will produce a laser beam of wavelength $9.6\mu\text{m}$. Normally $10.6\mu\text{m}$ transition is more intense than $9.6\mu\text{m}$ transition. The power output from this laser is 10kW.

Characteristics:

1. Type: It is a molecular gas laser.
2. Active medium: A mixture of CO_2 , N_2 and helium or water vapour is used as active medium
3. Pumping method: Electrical discharge method is used for Pumping action
4. Optical resonator: Two concave mirrors form a resonant cavity
5. Power output: The power output from this laser is about 10kW.
6. Nature of output: The nature of output may be continuous wave or pulsed wave.
7. Wavelength of output: The wavelength of output is $0.6\mu\text{m}$ and $10.6\mu\text{m}$.

Advantages:

1. The construction of CO_2 laser is simple
2. The output of this laser is continuous.
3. It has high efficiency
4. It has very high output power.
5. The output power can be increased by extending the length of the gas tube.

Disadvantages:

1. The contamination of oxygen by carbon monoxide will have some effect on laser action
2. The operating temperature plays an important role in determining the output power of laser.
3. The corrosion may occur at the reflecting plates.
4. Accidental exposure may damage our eyes, since it is invisible (infra red region) to our eyes.

Applications:

1. High power CO_2 laser finds applications in material processing, welding, drilling, cutting soldering etc.
2. The low atmospheric attenuation ($10.6\mu\text{m}$ makes CO_2 laser suitable for open air communication).
3. It is used for remote sensing.

4. It is used for treatment of liver and lung diseases.
5. It is mostly used in neuro surgery and general surgery.

Applications of Lasers:

Because of these unique properties, lasers are used in various applications.

The most significant applications of lasers include:

Lasers in Medicine

1. Lasers are used for bloodless surgery.
2. Lasers are used to destroy kidney stones.
3. Lasers are used in cancer diagnosis and therapy.
4. Lasers are used for eye lens curvature corrections.
5. Lasers are used in fiber-optic endoscope to detect ulcers in the intestines.
6. The liver and lung diseases could be treated by using lasers.
7. Lasers are used to study the internal structure of microorganisms and cells.
8. Lasers are used to produce chemical reactions.
9. Lasers are used to create plasma.
10. Lasers are used to remove tumors successfully.
11. Lasers are used to remove the caries or decayed portion of the teeth.
12. Lasers are used in cosmetic treatments such as acne treatment, cellulite and hair removal.

Lasers in Communications

1. Laser light is used in optical fiber communications to send information over large distances with low loss.
2. Laser light is used in underwater communication networks.
3. Lasers are used in space communication, radars and satellites.

Lasers in Industries

1. Lasers are used to cut glass and quartz.
2. Lasers are used in electronic industries for trimming the components of Integrated Circuits (ICs).
3. Lasers are used for heat treatment in the automotive industry.
4. Laser light is used to collect the information about the prefixed prices of various products in shops and business establishments from the bar code printed on the product.
5. Ultraviolet lasers are used in the semiconductor industries for photolithography. Photolithography is the method used for manufacturing printed circuit board (PCB) and microprocessor by using ultraviolet light.
6. Lasers are used to drill aerosol nozzles and control orifices within the required precision.

Lasers in Science and Technology

1. A laser helps in studying the Brownian motion of particles.
2. With the help of a helium-neon laser, it was proved that the velocity of light is same in all directions.
3. With the help of a laser, it is possible to count the number of atoms in a substance.
4. Lasers are used in computers to retrieve stored information from a Compact Disc (CD).
5. Lasers are used to store large amount of information or data in CD-ROM.
6. Lasers are used to measure the pollutant gases and other contaminants of the atmosphere.
7. Lasers helps in determining the rate of rotation of the earth accurately.
8. Lasers are used in computer printers.
9. Lasers are used for producing three-dimensional pictures in space without the use of lens.
10. Lasers are used for detecting earthquakes and underwater nuclear blasts.
11. A gallium arsenide diode laser can be used to setup an invisible fence to protect an area.

Lasers in Military

1. Laser range finders are used to determine the distance to an object.
2. The ring laser gyroscope is used for sensing and measuring very small angle of rotation of the moving objects.
3. Lasers can be used as a secretive illuminators for reconnaissance during night with high precision.
4. Lasers are used to dispose the energy of a warhead by damaging the missile.
5. Laser light is used in LIDAR's to accurately measure the distance to an object.

FIBER OPTICS

Introduction: In 1870 John Tyndall demonstrated that light follows the curve of a stream of water pouring from a container; it was this simple principle that led to the study and development of application of the fiber optics. The transmission of information over fibers has much lower losses than compared to that of cables. The optical fibers are most commonly used in telecommunication, medicine, military, automotive and in the area of industry. In fibers, the information is transmitted in the form of light from one end of the fiber to the other end with min. losses.

Principle of Optical Fiber

Optical fiber is a cylinder of transparent dielectric medium and designed to guide visible and infrared light over long distances. Optical fibers work on the principle of **total internal reflection**.

Optical fiber is very thin and flexible medium having a cylindrical shape consisting of three sections

- 1) The core material
- 2) The cladding material
- 3) The outer jacket

The structure of an optical is shown in figure.8.10. The fiber has a core surrounded by a cladding material whose reflective index is slightly less than that of the core material to satisfy the condition for total internal reflection. To protect the fiber material and also to give mechanical support there is a protective cover called outer jacket. In order to avoid damages there will be some cushion between cladding protective cover.

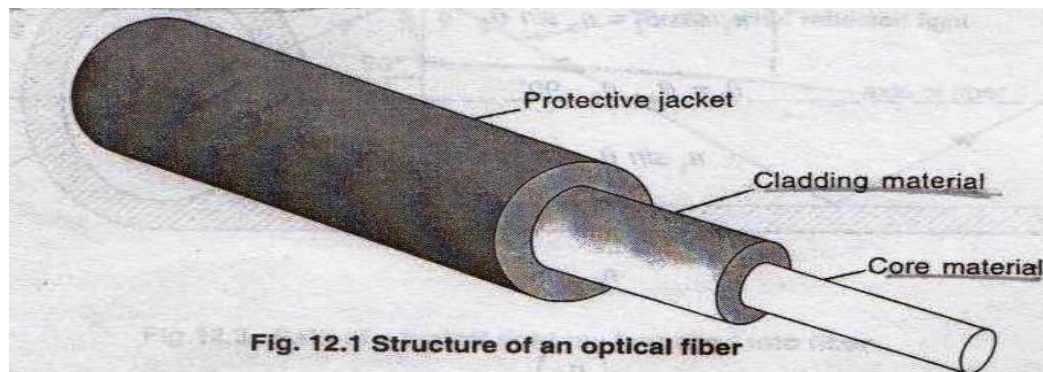


Fig. 1 Structure of an optical fiber

When a ray of light passes from an optically denser medium into an optically rarer medium the refracted ray bends away from the normal. When the angle of incidence is increased angle of refraction also increases and a stage is reached when the refracted ray just grazes the surface of separation of core and cladding. At this position the angle of refraction is 90 degrees. This angle of incidence in the denser medium is called the **critical angle (θ_c)** of the denser medium with respect to the rarer medium and is shown in the fig2. If the angle of incidence is further increased then the totally reflected. This is called total **internal reflection**. Let the refractive indices of core and cladding materials be n_1 and n_2 respectively.

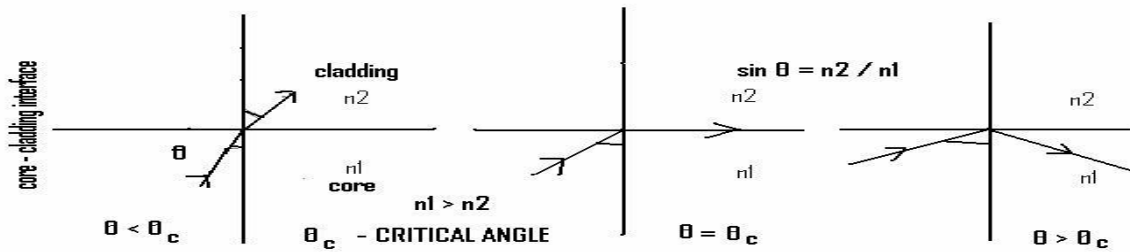


Fig.2 Total internal reflection.

When a light ray, travelling from an optically denser medium into an optically rarer medium is incident at angle greater than the critical angle for the two media. The ray is totally reflected back into the medium by obeying the law of reflection. This phenomenon is known as total internal reflection.

According to law of refraction,

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\text{Here } \theta_1 = \theta_c \quad \theta_2 = 90$$

$$n_1 \sin \theta_c = n_2 \sin(90)$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right) \quad \text{----- (1)}$$

Equation (1) is the expression for condition for total internal reflection. In case of total internal reflection, there is absolutely no absorption of light energy at the reflecting surface. Since the entire incident light energy is returned along the reflected light it is called total internal reflection.

reflection. As there is no loss of light energy during reflection, hence optical fibers are designed to guide light wave over very long distances.

Construction of optical fiber:

The optical fiber mainly consists of the following parts.

i .Core ii .Cladding iii .Silicon coating iv .Buffer jacketv .Strength material vi . Outer jacket

- A typical glass fiber consists of a central core of thickness $50\mu\text{m}$ surrounded by cladding.
- Cladding is made up of glass of slightly lower refractive index than core's refractive index, whose overall diameter is $125\mu\text{m}$ to $200\mu\text{m}$. Of course both core and cladding are made of same glass and to put refractive index of cladding lower than the refractive index of core, some impurities like Boron, Phosphorous or Germanium are doped.
- Silicon coating is provided between buffer jacket and cladding in order to improve the quality of transmission of light.
- Buffer jacket over the optical fiber is made of plastic and it protects the fiber from moisture and abrasion.
- In order to provide necessary toughness and tensile strength, a layer of strength material is arranged surrounding the buffer jacket.
- Finally the fiber cable is covered by black polyurethane outer jacket. Because of this arrangement fiber cable will not be damaged during hard pulling ,bending ,stretching or rolling, though the fiber is of brittle glass.



Fig. 28 Structure of an optical fiber.

Acceptance Angle and Acceptance Cone

Acceptance angle: Acceptance angle is the maximum angle of launch of light beam at its end of the fiber and to enable the entire light to propagate through the core. Fig.8.12 shows longitudinal cross section of the launch of a fiber with a ray entering it. The light is entered from a medium of refractive index n_0 (for air $n_0=1$) into the core of refractive index n_1 . The ray (OA) enters with an angle of incidence to the fiber end face i.e. the incident ray makes angle with the fiber axis which is nothing but the normal to the end face of the core. Let a right ray OA enters the fiber at an angle to the axis of the fiber. The end at which light enter the fiber is called the **launching pad**.

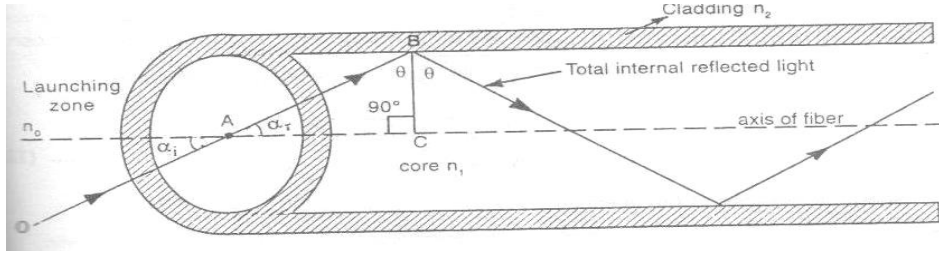


Fig.3 Path of atypical light ray launched into fiber.

Let the refractive index of the core be n_1 and the refractive index of cladding be n_2 . Here $n_1 > n_2$. The light ray reflects at an angle and strikes the core cladding interface at angle θ . If the angle θ is greater than its critical angle θ_c , the light ray undergoes total internal reflection at the interface.

According to Snell's law

$$n_0 \sin \alpha_i = n_1 \sin \alpha_r \quad \rightarrow (2)$$

From the right angled triangle ABC

$$\alpha_r + \theta = 90^\circ$$

$$\alpha_r = 90 - \theta \quad \rightarrow (3)$$

Substituting (3) in (2), we get

$$\begin{aligned} n_0 \sin \alpha_i &= n_1 \sin(90 - \theta) \\ &= n_1 \cos \theta \end{aligned}$$

$$\sin(\alpha_i) = \frac{n_1}{n_0} \cos \theta \quad \rightarrow (4)$$

When $\theta = \theta_c$ then $\alpha_i = \alpha_{max}$

$$\sin(\alpha_{max}) = \frac{n_1}{n_0} \cos \theta_c \quad \rightarrow (5)$$

From equation (1) $\sin \theta_c = \frac{n_2}{n_1}$

$$\cos \theta_c = \sqrt{1 - \sin^2 \theta_c} = \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2} = \frac{\sqrt{n_1^2 - n_2^2}}{n_1} \quad \rightarrow (6)$$

Substitute equation (6) in equation (5)

$$\sin(\alpha_{max}) = \left(\frac{n_1}{n_0}\right) \frac{\sqrt{n_1^2 - n_2^2}}{n_1} = \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \quad \rightarrow (7)$$

If the medium surrounding fiber is air, then $n_0 = 1$

$$\sin(\alpha_{max}) = \sqrt{n_1^2 - n_2^2} \quad \rightarrow (8)$$

This maximum angle is called the **acceptance angle or the acceptance cone half angle** of the fiber.

The **acceptance angle** may be defined as the maximum angle that a light ray can have with the axis of the fiber and propagate through the fiber. Rotating the acceptance angle about the

fiber axis (fig .8.13) describes the acceptance cone of the fiber. Light launched at the fiber end within this acceptance cone alone will be accepted and propagated to the other end of the fiber by total internal reflection. Larger acceptance angles make launching easier. Light gathering capacity of the fiber is expressed in terms of maximum acceptance angle and is termed as “Numerical Aperture”.

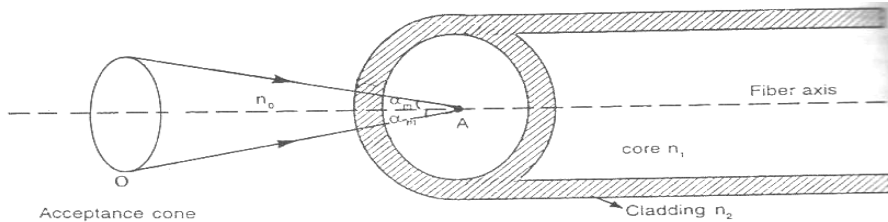


Fig.4 Acceptance cone

Numerical Aperture

Numerical Aperture of a fiber is measure of its light gathering power. The numerical aperture (NA) is defined as the Sin of the maximum acceptance angle.

$$\text{Numerical aperture (NA)} = \sin(\alpha_{max}) = \sqrt{n_1^2 - n_2^2} \quad \rightarrow (9)$$

$$= \sqrt{(n_1 - n_2)(n_1 + n_2)}$$

$$= \sqrt{((n_1 + n_2) n_1 \Delta)} \quad \rightarrow (10)$$

Where $\Delta = \frac{(n_1 - n_2)}{n_1}$ called as fractional differences in refractive indices n_1 and n_2 are the refractive indices of core and cladding material respectively.

As $n_1 \approx n_2$, we can take $n_1 + n_2 = 2n_1$

$$\text{Then numerical aperture} = \sqrt{2n_1^2 \Delta} = n_1 \sqrt{2\Delta} \quad \rightarrow (11)$$

Numerical aperture is a measure of amount of light that can be accepted by a fiber. From equation (9) it is seen that numerical aperture depends only on the refractive indices of core and cladding materials and it is independent on the fiber dimensions. Its value ranges from 0.1 to 0.5. A large NA means that the fiber will accept large amount of light from the source.

Step index fibers and Graded index fiber -transmission of signals in them:

Based on the variation of refractive index of core, optical fibers are divided into 2 types

They are (1) Step index fibers

(2) Graded index fibers

Again based on the mode of propagation, all these fibers are divided into two types.

They are (1) single mode

(2) multimode fibers.

In all optical fibers, the refractive index of cladding material is uniform. Now, we will see the construction, refractive index of core and cladding with radial distance of fiber, ray propagation and applications of above optical fibers.

1.Step index fiber: The refractive index is uniform throughout the core of this fiber. As we go radially in this fiber, the refractive index undergoes a step change at the core-cladding interface.

Based on the mode of propagation of light rays, step index fibers are of 2 types: a) single mode step index fiber & b) multimode step index fibers. Mode means, the number of paths available for light propagation of fiber. We describes the different types of fiber below

a) **Single mode step index fiber:** The core diameter of this fiber is about 2 to 8 μm and outer diameter of cladding is 60 to 70 μm . There is only one path for ray propagation. So, it is called **single mode fiber**. The cross sectional view, refractive index profile and ray propagation are shown in fig.a . In this fiber, the transmission of light is by successive total internal reflections i.e. it is a reflective type fiber. Nearly 80% of the fibers manufactured today in the world are single mode fibers. So, they are extensively used.

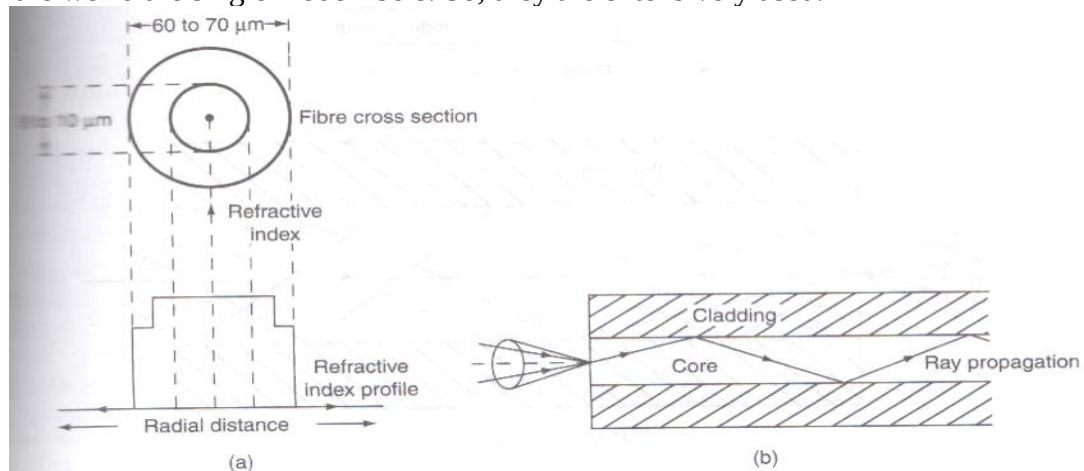


Fig .Single mode step index fiber ;(a) Cross sectional view and refractive index profile ;(b) Ray propagation

Multimode step index fiber: The construction of multimode step index fiber is similar to single mode step index fiber except that its core and cladding diameters are much larger to have many paths for light propagation. The core diameter of this fiber varies from 50 to 200 μm and the outer diameter of cladding varies from 100 to 250 μm . Light propagation in this fiber is by multiple total internal reflections i.e it is a reflective type fiber.

Transmission of signal in step index fiber: Generally the signal is transmitted through the fiber in digital form i.e. in the form of 1's and 0's. The propagation of pulses through the multimode fiber is shown in fig8.13 (i)(b). The pulse which travels along path 1(straight) will reach first at the other end of fiber. Next the pulse that travels along with path 2(zig-zag) reaches the other end. Hence, the pulsed signal received at the other end is broadened. This is known as intermodal dispersion. This imposes limitation on the separation between

pulses and reduces the transmission rate and capacity. To overcome this problem, graded index fibers are used.

2) **Graded index fiber:** In this fiber, the refractive index decreases continuously from center radially to the surface of the core. The refractive index is maximum at the center and minimum at the surface of core. This fiber can be single mode or multimode fiber. The cross sectional view, refractive index profile and ray propagation of multimode graded index fiber are shown in fig.8.13(ii)(a). The diameter of core varies from 50 to 200 μm and outer diameter of cladding varies from 100 to 250 μm .

The refractive index profile is circularly symmetric. As refractive index changes continuously radially in core, light rays suffer continuous refraction in core. The propagation of light ray is not due to total internal reflection but by refraction as shown in fig.8.13 (ii). (b). in graded index fiber, light rays travel at different speed in different paths of the fiber. Near the surface of the core, the refractive index is lower, so rays near the outer surface travel faster than the rays travel at the center. Because of this, all the rays arrive at the receiving end of the fiber approximately at the same time. This fiber is costly. .

Transmission of signal graded index fiber: In multimode graded index fiber, large number of paths is available for light ray propagation. To discuss about inter modal dispersion, we consider ray path 1 along the axis of fiber.

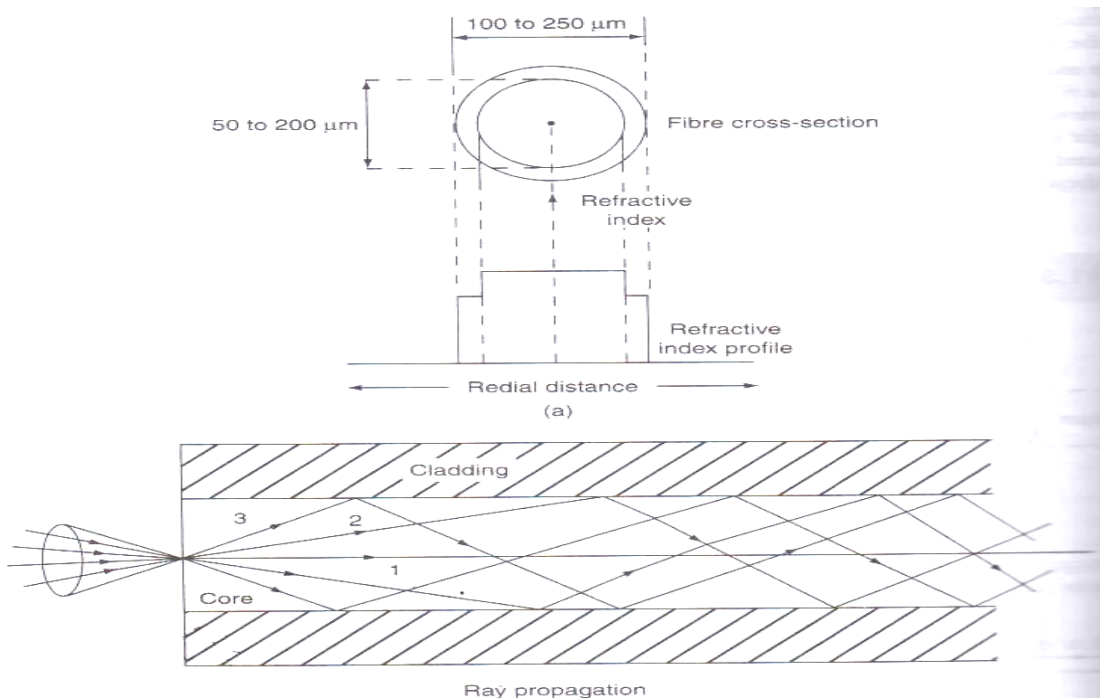


Fig.8.13(ii).Multimode step index fibre(a)Cross sectional view and refractive index profile(b)Ray propagation

As shown in fig.8.13. (ii)(b) and another ray path 2. Along with the axis of fiber, the refractive index of core is maximum, so the speed of ray along path 1 is less. Path 2 is sinusoidal and it is

longer, along this path refractive index varies. The ray mostly travels in low refractive region, so the ray 2 moves slightly faster. Hence, the pulses of signals that travel along path 1 and path 2 reach other end of fiber simultaneously. Thus, the problem of intermodal dispersion can be reduced to a large extent using graded index fiber.

Differences between step index fiber and graded index fibers:

Step index fiber	Graded index fiber
1.The refractive index of core Is uniform and step or abrupt Change in refractive index Takes place at the interface of core and cladding in step index fibers	1.The refractive index of core is non uniform, the refractive index of core decreases Parabolicity from the axis of the fiber to its surface.
2. The light rays propagate in zigzag manner inside the core. The rays travel in the fiber as meridional rays they cross the fiber axis for every reflection.	2. Light rays propagate in the form of skew rays or helical rays. They will not cross the fiber axis

Difference between single mode fibers and multi mode fibers:

Single mode fiber	Multimode fiber
1. In single mode fiber there is only one path for ray propagation	1. In multimode fiber, large number of paths is available for light ray propagation.
2. A single mode step index fiber has less core diameter ($< 10 \mu\text{m}$) and the difference between the reflective indices of core and cladding is very small.	2. Multi mode fibers, large number of paths are available for light ray propagation.
3. In single mode fibers, there is no dispersion.	3. There is signal distortion and dispersion takes place in multimode fibers.
4. The band width is about 50 MHz for multimode step index fiber where as it is more than 1000 MHz km in case of single mode step index fiber.	4. The band width of the fiber lies in between 200 MHz km to 600 MHz km even though theoretically it has an infinite bandwidth.
5. NA of multimode step index fiber is more where as in single mode step index fibers, it is very less.	5. NA of graded index fibers is less.
6. Launching of light into single mode fibers is difficult.	6. Launching of light into multimode fibers is easy.
7. Fabrication cost is very high.	7. Fabrication cost is less

V-Number: The V-number or normalized frequency parameter gives the upper limit of the number of modes that can be transmitted in a multimode optical fiber. It depends on the core diameter, d_c , NA and Wavelength. When a beam of monochromatic rays are launched at the end of fiber with a angle less than the acceptance angle also may not propagate through the fiber. In practice a limited no. of modes of propagation are possible. The reason for limited mode acceptance in fiber is Phase change occurrence when the light ray travels forward. The phase shift is due to 1.Optical path length 2.Total internal reflection at the core-cladding interface.Each mode has a definite value of V-number below which the mode is cut off.

$$V = \frac{\pi d}{\lambda_0} \sqrt{n_1^2 - n_2^2}$$

Where V- V-number,

d- diameter of core,

λ_0 – wavelength of light,

n_1 – Refractive index of core,

n_2 – Refractive index of cladding.

V-number in terms of Numerical Aperture $V = \frac{\pi d}{\lambda_0} (NA)$

$$V = \frac{\pi d}{\lambda_0} n_1 \sqrt{2 \Delta}$$

The maximum no. of modes supported by a Step Index fiber is given by $N_m = \frac{V^2}{2}$.

The maximum no. of modes supported by a Graded Index fiber is given by $N_m = \frac{V^2}{4}$.

Attenuations in optical fiber (or losses):

While transmitting the signals through optical fiber some energy is lost due to few reasons.

The major losses in fibers are

1. Distortion losses
2. Transmission losses
3. Bending losses.

1. Distortion losses:-

When a pulse is launched at one end of the fiber and collected at the other end of the fiber, the shape and size of the pulse should not be changed. Distortion of signals in optical fiber is an undesired feature. If the out put pulse is not same as the input pulse, then it is said that the pulse is distorted .If the refractive index of the core is not uniform most of the rays will travel through the medium of lower refractive index region. Due to this the rays which are travel in fiber will become broadened. Because of this the out put pulses will no longer matches with the input pulses. The distortion takes place due to the presence of imperfections, impurities and

doping concentrations in fiber crystals. Distortion is large in multi mode than in single mode fiber.

2. Transmission losses (attenuation):-

The attenuation or transmission losses may be classified into two categories

- i) Absorption losses
- ii) Scattering losses

i) Absorption losses:-

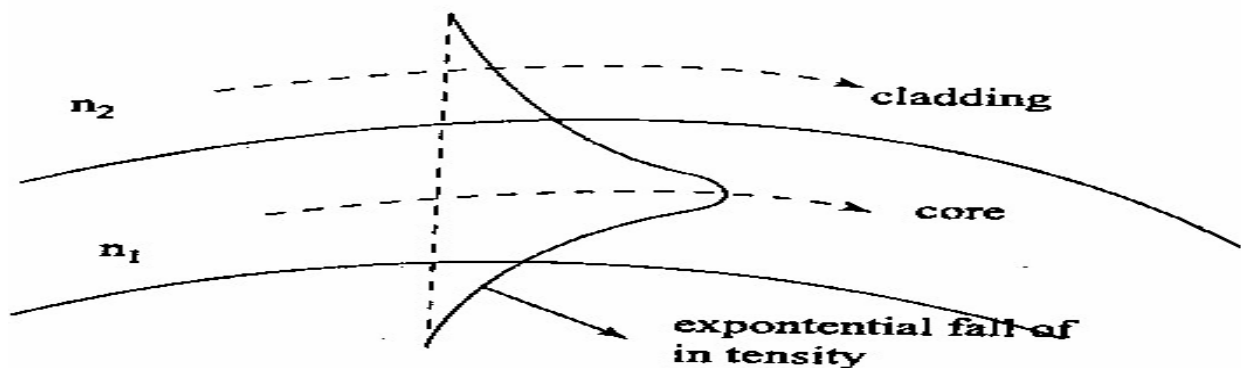
Absorption is a characteristic possessed by all materials every material in universe absorb suitable wavelengths as they incident or passed through the material. Three different mechanisms contribute to absorption losses in glass fiber. These are Ultraviolet absorption, IR absorption and Ion resonance absorption. In the same way core material of the fiber absorbs wavelengths as the optical pulses or wavelengths pass through it.

ii) Scattering losses:-

The core medium of the fiber is made of glass or silica .In the passage of optical signals in the core medium if crystal defects are encountered, they deviate from the path and total internal reflection is discontinued, hence such signals will be destroyed by entering into the cladding however attenuation is minimum in optical fibers compared to other cables.

ii) Bending losses:-

The distortion of the fiber from the ideal strait line configuration may also result in fiber .Let us consider a way front that travels perpendicular to the direction of propagation. In order to maintain this, the part of the mode which is on the outside of the bend has to travel faster than that on the inside. As per the theory each mode extends an infinite distance into the cladding though the intensity falls exponentially. Since the refractive index of cladding is less than that of the core ($n_1 > n_2$), the part of the mode traveling in the cladding will attempt to travel faster. As per Einstein's theory of relativity since the part of the mode cannot travel faster the energy associated with this particular part of the mode is lost by radiation.



Attenuation loss is generally measured in terms of decibels (dB), which is a logarithmic unit.

$$\text{Loss of optical power} = -10 \log (P_{\text{out}} / P_{\text{in}}) \text{ dB}$$

Where P_{out} is the power emerging out of the fiber.

P_{in} is the power launched into the fiber.

Application of Fiber optic sensors and detector:

Sensors are devices used to measure or monitor quantities such as displacement pressure, temperature, flow rate liquid level, chemical composition etc. A smoke detector and pollution detector can be made from fibers. There are two types of fiber optics sensors.

1. Intrinsic or Active sensors:

Here guided like in the fiber gauge modulated by the variable to be measured and then de modulated. Thus sensing and simultaneous measurements of taking place in the active sensors. They are

i. Intensity modulate sensors:

Which are based on the change in absorption or transmission of light, refractive index, temperature etc.

ii. Phase modulated sensors:

Which involve the interference between signal and reference in the interferometer leading to a shift in the interference fringes. These will have high sensitivity.

iii. Polarization modulated sensors:

Which involve the change in polarization state of the guided signal by the variable.

2. Extrinsic or Passive sensors:

Here the modulation takes place outside the optical fiber which acts merely as a convenient transmission channel for the light radiation. It has a sensor head and the sensed optical signal is transferred to remote point for signal processing. Based on the parameters different fiber sensors are prepared. For ex: micro bending displacement sensor, laser Doppler velocity sensor, fiber optic sensor for the measurement of electric field or voltage, faraday reaction sensor based on the measurement of magnetic field. Fiber optic temperature sensors are used to measure temperature differences of the order 0.001°C with much low response time.

3. Optical Communications: In recent years optical fibers are being used instead of copper cables, where light signals replace electrical signals.

4. Medical Field: The fiber optic technology is used in medical diagnostics and medical procedures. It is used in Endoscopy, Cardiology, Ophthalmology and treatment of Cancer.

5. Military Field: Fiber guided missiles are used in recent wars.

6. Lighting and Decorations:

The use of fiber optics in the area of decorative illumination has also grown over the years. Fiber optic cables provide an easy, economical and attractive solution to lighting projects. As a result, they are widely used in lighting decorations and illuminating Christmas trees.

7. Mechanical Inspections:

Fiber optic cables are widely used in the inspection of hard-to-reach places. Some such applications are on-site inspections for engineers and also inspection of pipes for plumbers.

8. Automotive Industry:

Fiber optic cables play an important role in the lighting and safety features of present day automobiles. They are widely used in lighting, both in the interior and exterior of vehicles.